

Maincraft Technology

AIML Internship Task 3

Model Validation, Overfitting Control & Hyperparameter Tuning

Submitted By:

Sampada Bandre

Internship Domain:

Artificial Intelligence & Machine Learning

Introduction

The objective of this project was to improve the reliability and performance of a machine learning model for house price prediction using the California Housing Dataset. In real-world machine learning applications, a model must not only perform well on training data but also generalize effectively to unseen data. Therefore, techniques such as overfitting analysis, cross-validation, and hyperparameter tuning were applied to build a more robust predictive model.

The California Housing Dataset contains various housing-related features, including median income, house age, average rooms, average bedrooms, population, occupancy, latitude, and longitude. The target variable used for prediction was the median house value.

Methodology

The dataset was loaded using the Scikit-Learn library and explored through basic data analysis techniques. Missing value analysis confirmed that the dataset contained no null values. A correlation heatmap was generated to understand relationships between features and the target variable.

The target variable (MedHouseVal) was separated from the input features. Feature scaling was performed using StandardScaler to ensure that all features were on a similar scale, improving model performance and stability.

The dataset was divided into training and testing sets using an 80:20 ratio. Three machine learning algorithms were used for model comparison:

- Linear Regression
- Ridge Regression
- Decision Tree Regressor

Overfitting Analysis

Overfitting occurs when a model learns the training data too closely and performs poorly on unseen data. To analyze overfitting, a Decision Tree Regressor was trained and evaluated on both training and testing datasets.

The training RMSE and testing RMSE values were compared. A significant difference between these values indicates that the model may be overfitting the training data. This analysis helped determine whether additional optimization techniques were required to improve generalization performance.

Cross-Validation

Cross-validation was applied using 5-Fold Cross Validation. This technique divides the dataset into multiple subsets and evaluates the model several times using different train-test combinations.

Cross-validation provides a more reliable estimate of model performance compared to a single train-test split. The average cross-validation score was calculated to measure how consistently the model performs across different subsets of data.

Hyperparameter Tuning

Hyperparameter tuning was performed using GridSearchCV. Several combinations of Decision Tree parameters such as maximum tree depth, minimum samples split, and minimum samples leaf were tested automatically.

GridSearchCV selected the parameter combination that achieved the best cross-validation performance. The optimized model was then used for final evaluation.

This process helped reduce overfitting and improved the model's ability to generalize to unseen data.

Model Performance Comparison

The performance of all models was evaluated using Root Mean Squared Error (RMSE) and R^2 Score.

Model	RMSE	R^2 Score
Tuned Decision Tree	0.638936	0.688464
Ridge Regression	0.745554	0.575819
Linear Regression	0.745581	0.57578

The comparison table demonstrates the effectiveness of model optimization techniques. The model with the lowest RMSE and highest R^2 Score was selected as the final model.

Conclusion

This project successfully demonstrated model validation, overfitting analysis, cross-validation, and hyperparameter tuning using the California Housing Dataset. Cross-validation provided a more reliable measure of model performance, while GridSearchCV helped identify the optimal parameters for the Decision Tree Regressor.

The final model achieved improved predictive performance and better generalization capability compared to the baseline model. This task highlighted the importance of validation and optimization techniques in developing accurate and reliable machine learning solutions.

Overall, the project provided valuable hands-on experience in advanced machine learning workflows and model improvement strategies.